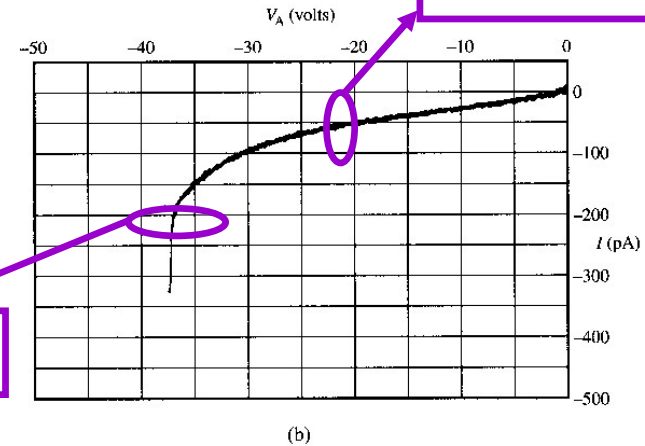
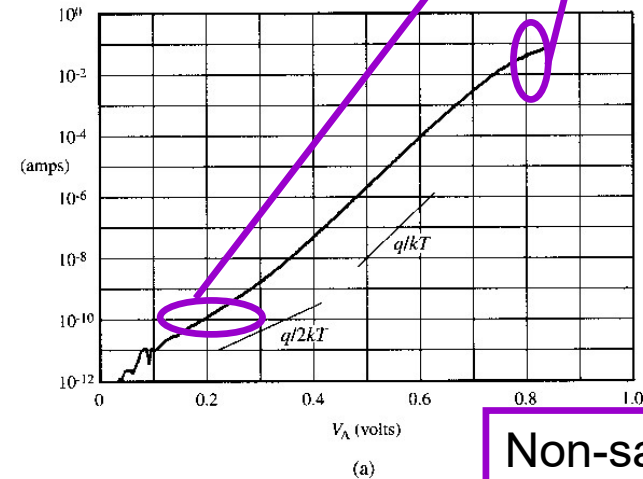
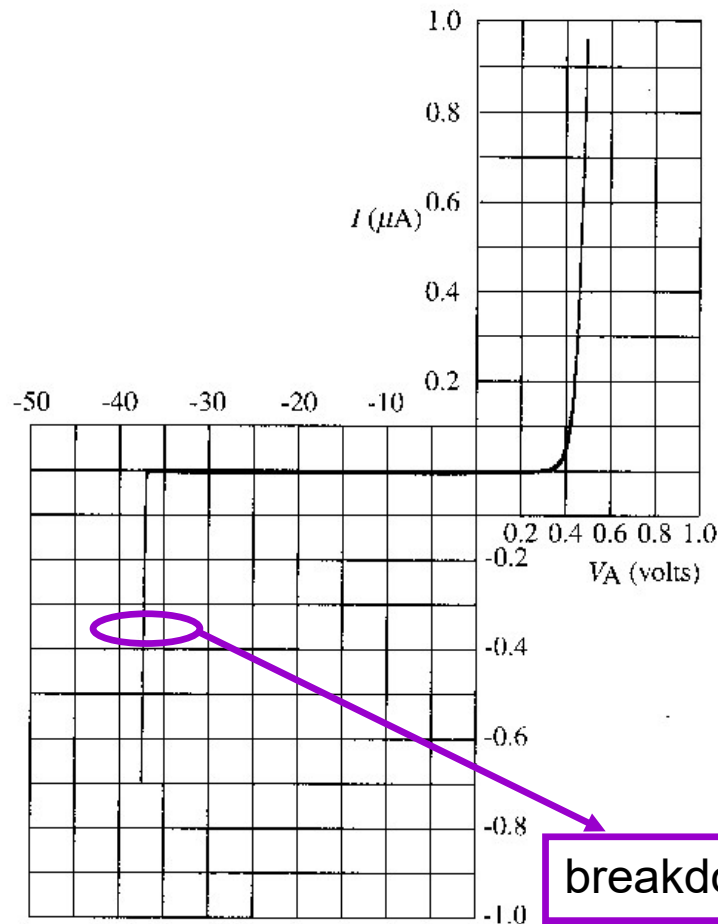


Current-Voltage Characteristics: Non-Ideal Effects

□ Experimental Current-Voltage Characteristics



Breakdown

□ Breakdown Phenomenal

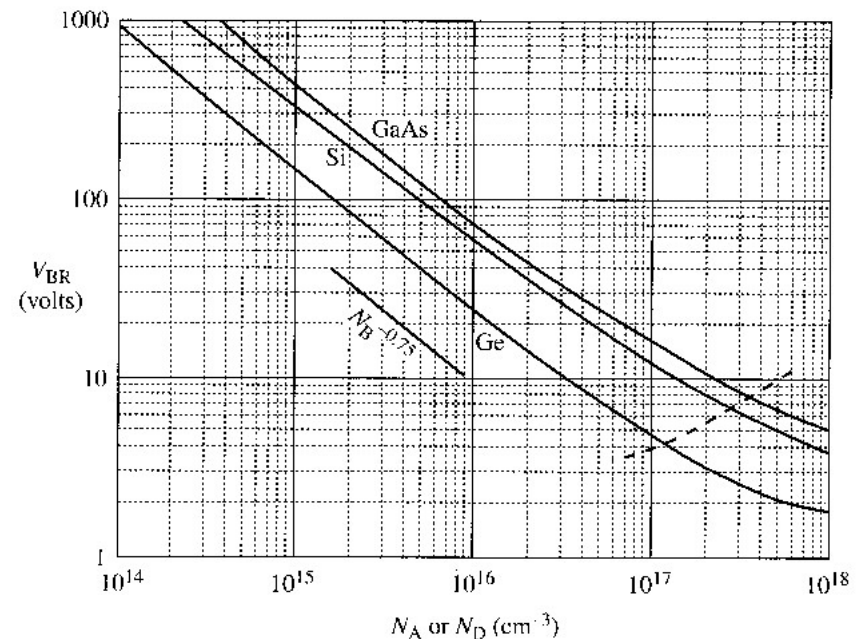
- “Breakdown” is referred to event when very large reverse current flows at a reverse voltage exceeding a certain value.

- Breakdown is a completely reversible process. That means breakdown does not damage the diode.

- The absolute value of the reverse voltage where the current goes to infinity is known as breakdown voltage (V_{BR}).

- Practically V_{BR} is measured where current exceeds a certain value, for example 1uA or 1mA.

- Normally, two kinds of mechanisms exist: Avalanche breakdown and Zener breakdown.

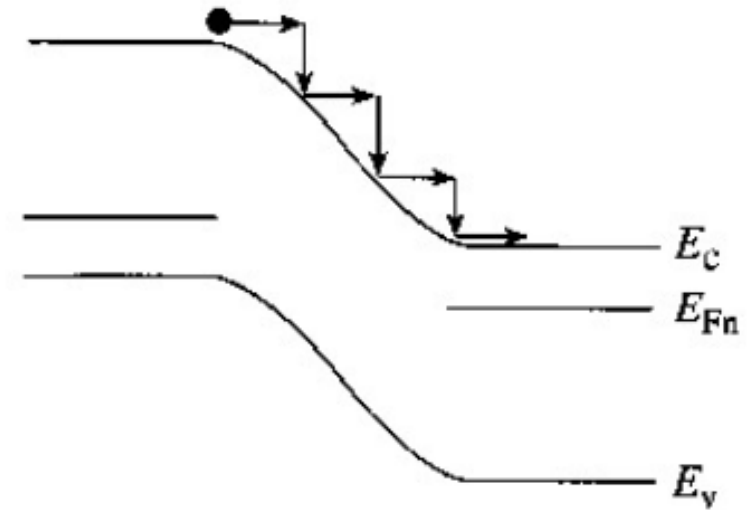


Breakdown

❑ Breakdown Phenomena: Avalanche Breakdown

Small Reverse Voltage (far below breakdown)

- The reverse current is due to minority carrier randomly entering the depletion region and being accelerated by the electric field to the other side.
- In crossing the depletion region, the carrier acceleration is not continuous but is interrupted by energy-losing collision with the semiconductor lattice.
- At small reverse bias, the energy lost by carriers per collision is relative small.
- The energy transferred to the lattice simply causes lattice vibrations. There is just localized heating that is readily dissipated.



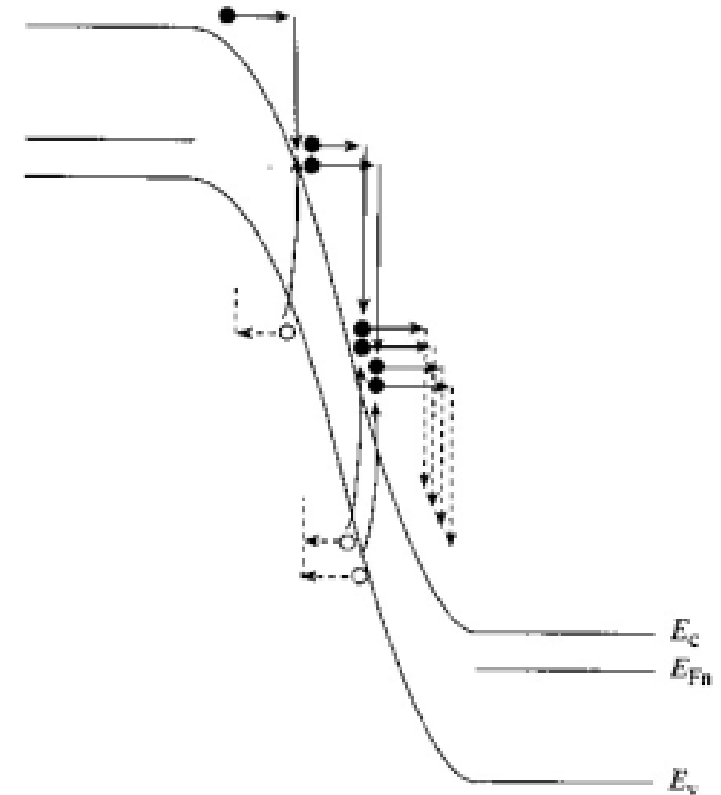
(a) Small reverse bias

Breakdown

□ Breakdown Phenomenon: Avalanche Breakdown

Large Reverse Voltage

- Approaching the breakdown voltage, the energy transferred per collision becomes sufficient to ionize a semiconductor atom.
- 'Ionize' means that the collision frees a valence electron from the atom, or causes an electron from the valence band to jump into the conduction band, thereby creating an electron-hole pair. (called **impact ionization**)
- The added carriers are immediately accelerated by the electric collisions and create even more carriers.
- The result is a snowballing creation of carriers very similar to an avalanche of snow on a mountain side.
- At the breakdown voltage, the carrier creation and reverse current effectively goes to infinity.



(b) $|V_A| \rightarrow V_{BR}$

Breakdown

□ Breakdown Phenomenal: Avalanche Breakdown

Dopant concentration dependence

- For avalanche breakdown, considering a step junction, the breakdown voltage is related with doping concentrations as follow

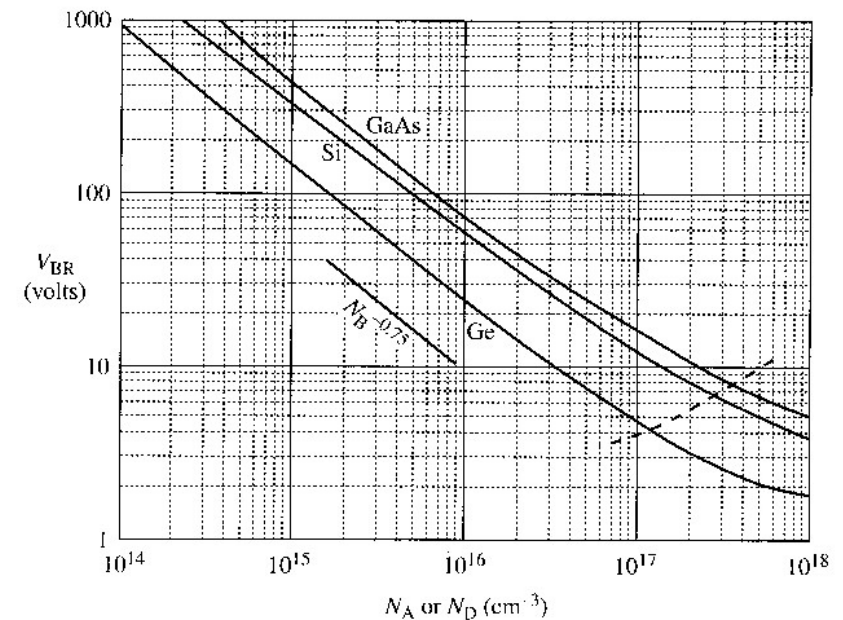
$$V_{BR} \propto \frac{N_A + N_D}{N_A N_D}$$

- For asymmetrically doped junctions

$$V_{BR} \propto \frac{1}{N_B}$$

where N_B is doping concentration of low side.

- Physical view:
Lower doping leads to a wider depletion width, which can support more voltage and hence a higher breakdown voltage.



Breakdown

□ Breakdown Phenomenal: Avalanche Breakdown

Temperature dependence

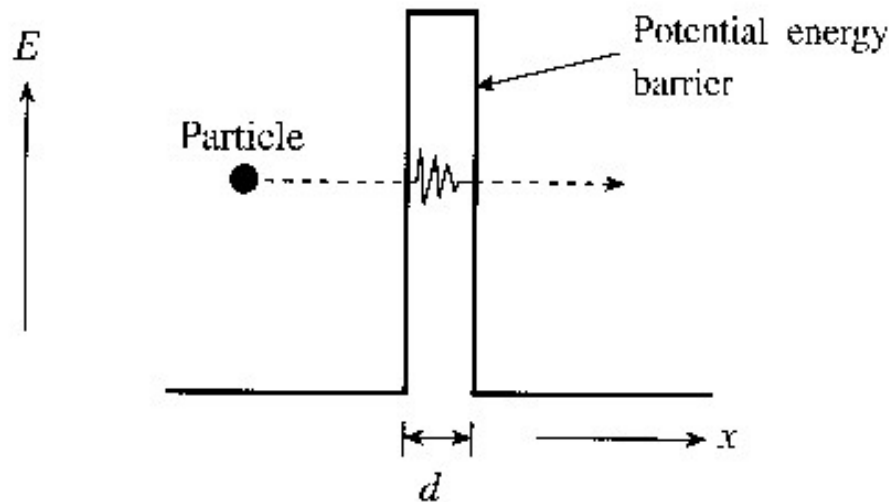
- The V_{BR} due to avalanching is found to **increase** as the **temperature** goes up.
- Lattice scattering increases as the temperature increases.
- Increasing lattice scattering means a **smaller mean free path**. Therefore, a larger electric field is needed for avalanching, and hence a higher breakdown voltage.

Breakdown

□ Breakdown Phenomenal: Zener Breakdown

Zener breakdown – a result of “tunneling”

- Zener process in the name given to the occurrence of “tunneling” in a reverse-biased diode.
- Tunneling is a phenomenon we have encountered that is of a purely **quantum-mechanical** (QM) nature. It has no classic analog.

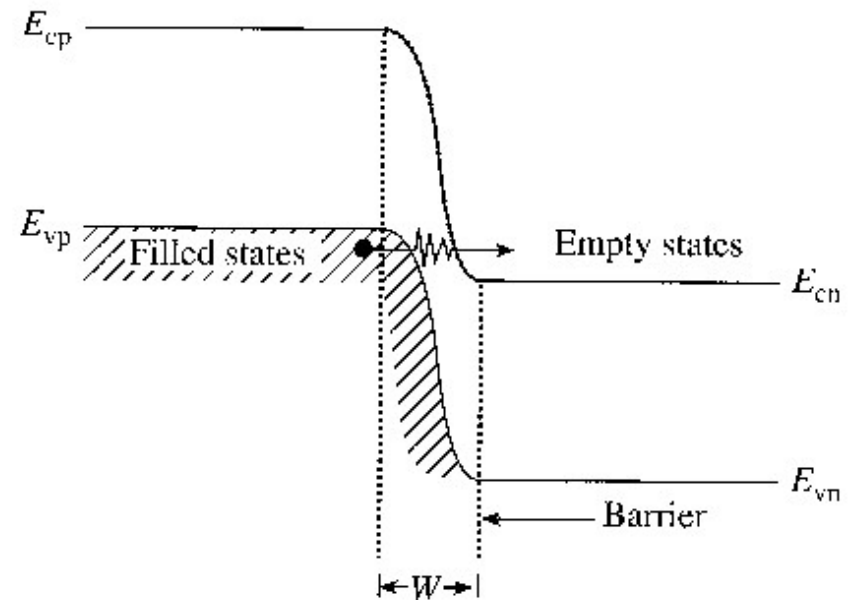


Breakdown

□ Breakdown Phenomenal: Zener Breakdown

There are two major requirements for tunneling to occur and be significant

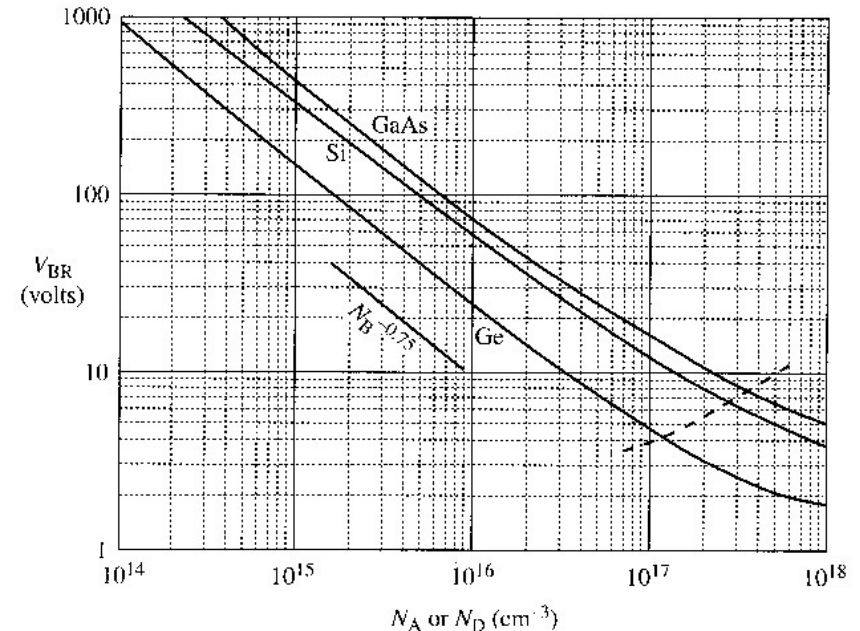
- (1) There must be filled states on one side of the barrier and empty states on the other side of the barrier at the same energy. **Tunneling cannot take place into a region void of allowed states.**
- (2) The width of potential energy barrier d (or depletion width W) must be **very thin**. QM tunneling becomes significant normally when $W < 10$ nm.



Breakdown

□ Breakdown Phenomenal: Zener Breakdown

- The greater the reverse bias, the larger the number of filled valence electron states on the p-side placed above the empty conduction band states on the n-side, and hence the greater the reverse-bias tunneling current.
- For tunneling to be significant, the barrier thickness (roughly the depletion width of pn junction), must be less than 10 nm. So, high doping is always needed to achieve such a low depletion width.



Thus, the breakdown voltage is always small.

Breakdown

□ Breakdown Phenomenal: Comparison of Avalanche and Zener Breakdown

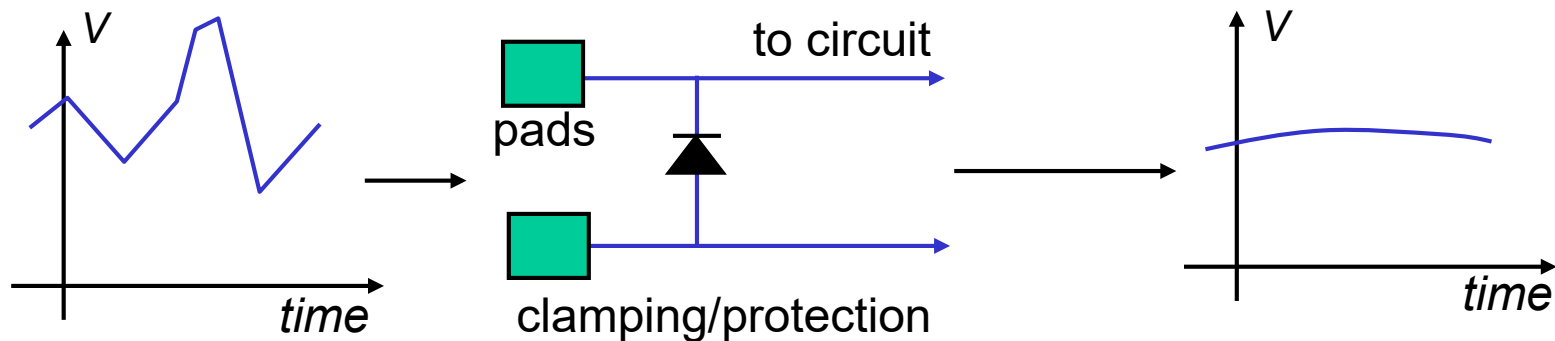
- Zener process always dominates in diodes where $V_{BR} < 4 E_g/q$ (~4.5 V in Si at 300 K)
- Avalanche always dominates in diodes where breakdown voltage is higher.
- The breakdown voltage associated with avalanching **increases** with **increasing temperature**, whereas the V_{BR} associated with Zener process **decreases** with **increasing temperature**.
- The breakdown characteristics associated with the Zener process are very “soft” compared to that associated with avalanching.

Breakdown

□ Breakdown Phenomenal: Example of Application

Application of diode breakdown

- Clamping devices for voltage stabilization and protection
- Question: can you use the forward bias region as clamping devices?



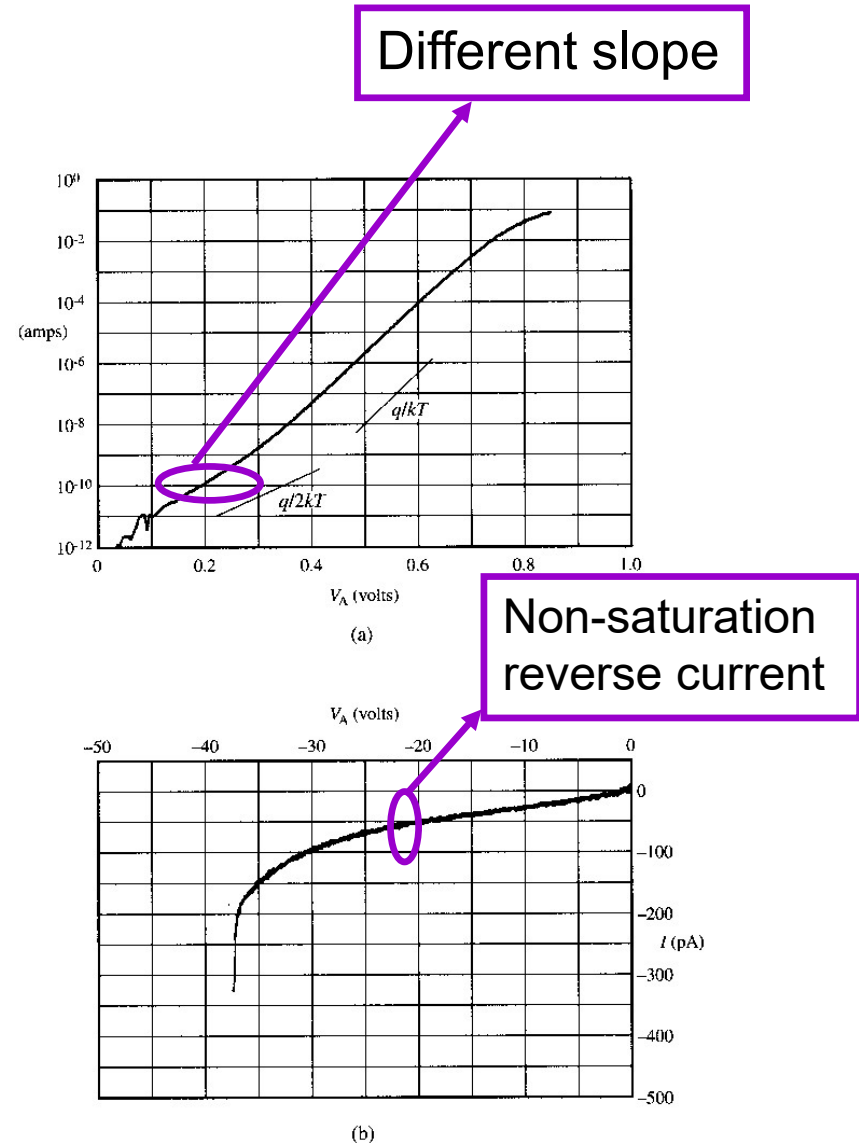
Recombination-Generation

□ Recombination-Generation

R-G current:

Refers to current that arises from thermal carrier recombination-generation in the depletion region that was assumed to be negligible in the derivation of ideal diode equation.

- (1) Non-saturation reverse current
- (2) Different slope in small forward bias region

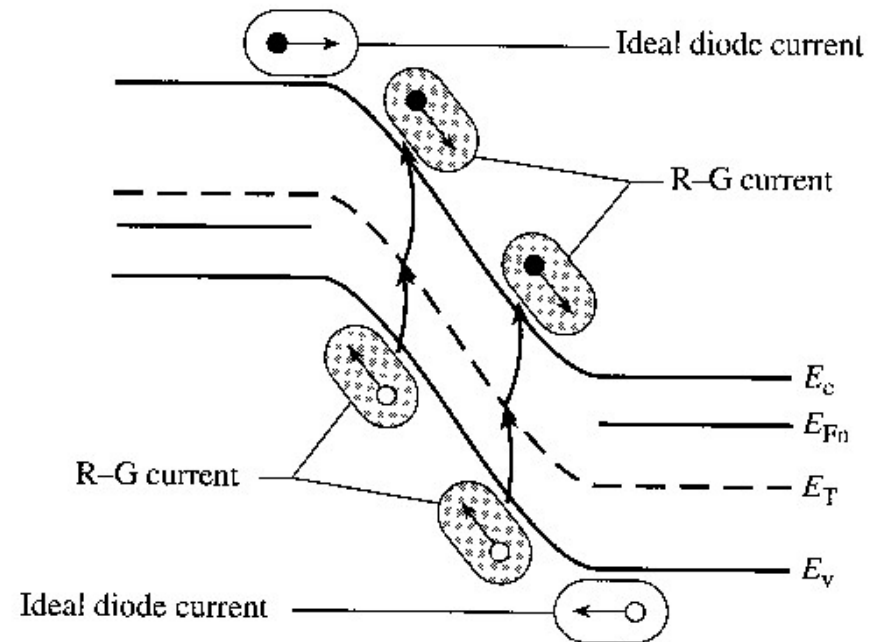


Recombination-Generation

□ Recombination-Generation

Reverse Bias

- In reverse bias, the carrier concentrations in the depletion region are reduced below their equilibrium values, leading to the thermal **generation of electrons and holes** throughout the region.
- The large electric field in the depletion region rapidly sweeps the generated carriers into the quasi-neutral regions, thereby adding to reverse current.



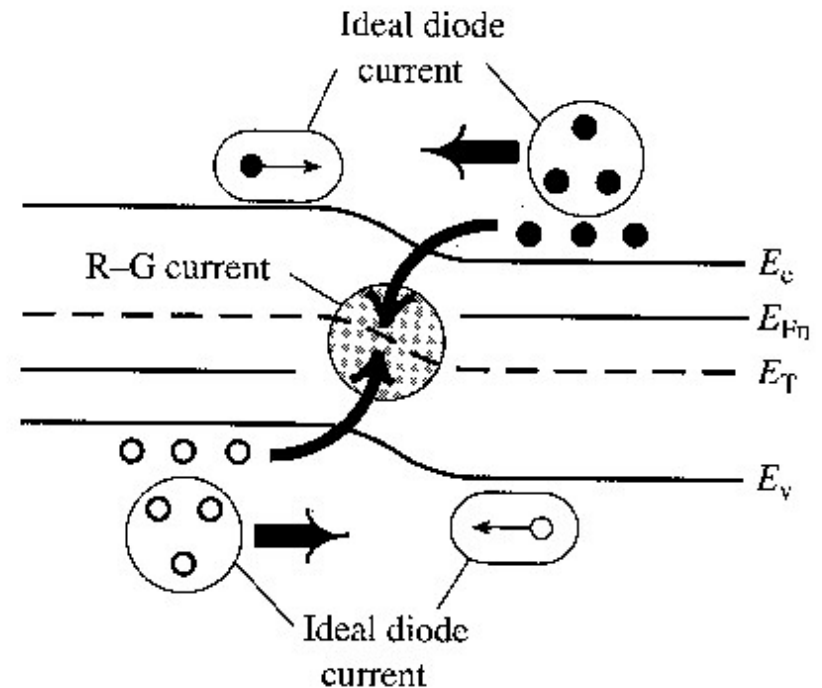
(a) Reverse Bias

Recombination-Generation

□ Recombination-Generation

Forward Bias

- In forward bias, the carrier concentrations **increase** in the depletion region above **their equilibrium values**, which gives rise to carrier recombination in the region.
- Thus, one can view that the added forward current as arising from the carriers that cannot make it over the potential hill being partially eliminated via recombination at R-G centers in the depletion region.



(b) Forward Bias

Recombination-Generation

□ Recombination-Generation

Calculation of R-G Currents

Step 1: Summing either electrons or holes created/destroyed in depletion region

$$I_{R-G} = -qA \int_{-X_p}^{X_n} \left. \frac{\partial n}{\partial t} \right|_{R-G} dx$$

Step 2: recall

$$\left. \frac{\partial n}{\partial t} \right|_{R-G} = - \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

Step 3: the R-G current can be given as below

$$I_{R-G} = qA \int_{-X_p}^{X_n} \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)} dx$$

Note: the net R-G rate is the same for electrons and holes under steady state conditions.

Recombination-Generation

□ Recombination-Generation

Calculation of R-G Currents

Reverse Bias

Step 4: In reverse bias, the electron and hole concentrations inside depletion can be ignored, so R-G current can be easily calculated as:

$$I_{R-G} = -\frac{qAn_i}{2\tau_0}W$$

where

$$\tau_0 \equiv \frac{1}{2} \left(\tau_p \frac{n_1}{n_i} + \tau_n \frac{p_1}{n_i} \right) = \frac{1}{2} \left(\tau_p e^{(E_T - E_i)/kT} + \tau_n e^{(E_i - E_T)/kT} \right)$$

Recombination-Generation

□ Recombination-Generation

Calculation of R-G Currents

Forward Bias

Step 4: For forward biases, the carrier concentrations cannot be neglected. Below gives a good estimation

$$I_{R-G} = \frac{qAn_i}{2\tau_0} W \frac{(e^{qV_A/kT} - 1)}{(1 + \frac{V_{bi} - V_A}{kT/q} \frac{\sqrt{\tau_n \tau_p}}{2\tau_0} e^{qV_A/2kT})}$$

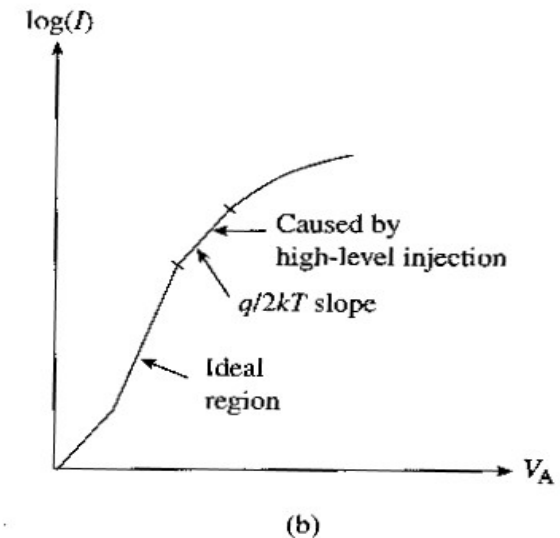
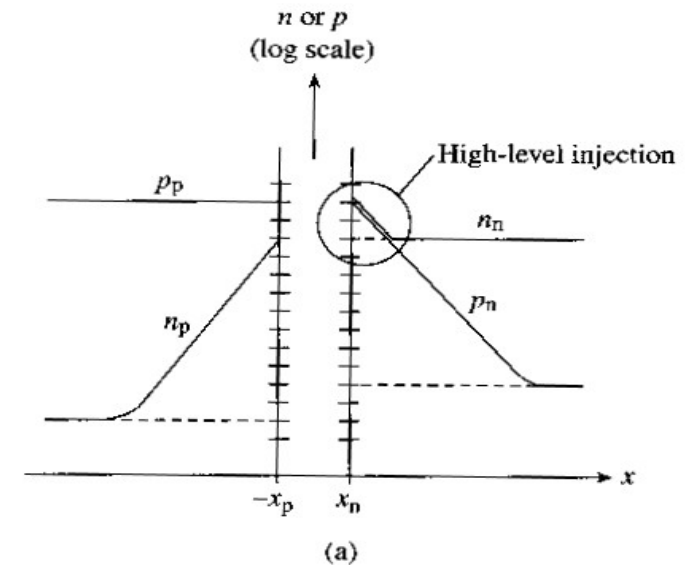
Step 5: The total current

$$I = I_{diode,ideal} + I_{R-G}$$

Series Resistance

High Current Phenomenon: High Level Injection

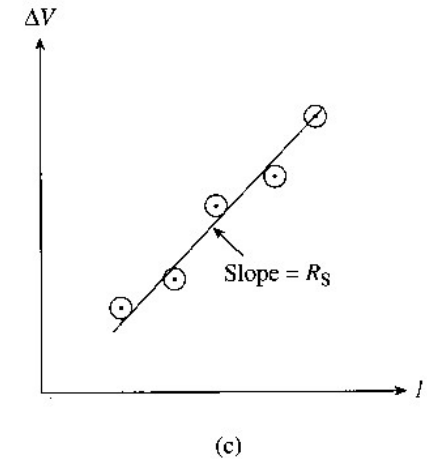
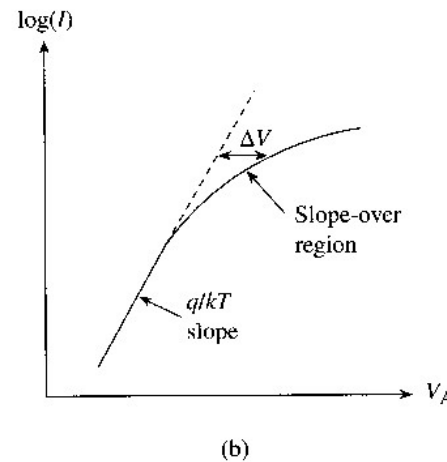
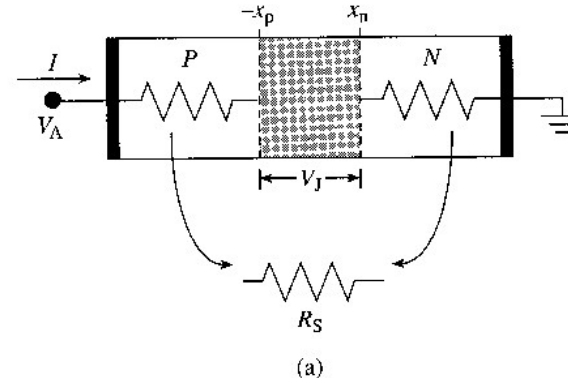
- The low-level injection begins to fail when the minority carrier concentration at the depletion region edge on the lightly doped side approaches the doping concentration.
- In Si at room temperature, this typically happens at applied voltage a few tenths of a volt below V_{bi} and further increased voltage leads to high-level injection.
- Under high-level injection, both the minority and majority carrier concentration adjacent to the depletion region are perturbed.
- The majority carrier concentration must increase to maintain approximate charge neutrality in the quasi-neutral regions.
- Detail analysis of high-level injection leads to a predicted current varying roughly as $\exp(q/2kT)$.



Series Resistance

□ High Current Phenomenal: Series Resistance

- The quasi-neutral regions have an inherent resistance determined by the doping and dimensions of the regions.
- These form a resistance R_S , in series with the current flow across the junction.
- At low current levels, the voltage drop across the series resistance, IR_S , is totally negligible compared to the applied voltage across the depletion region (or we call junction voltage, V_J). Thus, $V_J = V_A$.



Series Resistance

□ High Current Phenomenal: Series Resistance

- At current levels where IR_S becomes comparable to V_A , the applied voltage dropping appearing across the depletion region is reduced to

$$V_J = V_A - IR_S$$

- Effectively, part of the applied voltage is wasted.
- To correct for the series resistance, one merely replaces V_A by

$$V_J = V_A - IR_S$$

$$(\text{or } V_A \propto IR_S)$$

in previously derived I - V_A relationships.

